Decay & Scattering

Now that we know how to compute S-matrix elements, we want to relate them to physical, measurable quantities. The two most basic such quantities of particle physics experiments are decay rates & scattering cross sections.

With these in hand, we'll take a brief detour to discuss observables and statistics at collider physics experiments, as an example.

First, let us formally define a decay rate, as it requires filling one conceptual gap from our study of interacting ϕ^4 theory.

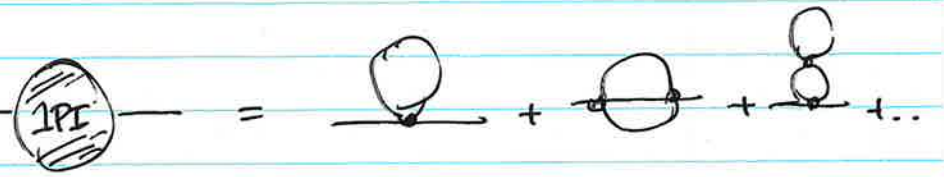
In ϕ^4 theory, we (reasonably!) assumed that we could consider asymptotic states related to free ~~particle~~ theory, single particle states. In general, this is not strictly true, as many (most!) of the particles in the SM are unstable: interactions allow them to decay to (energetically allowed) lighter, multi-particle states (as long as they conserve all other quantum numbers too).

The rate at which ~~these~~ these decays happen, or inverse lifetime, is the decay rate we wish to compute.

But how do we compute the rate, and how is this defined in QFTs?

Fortunately, our prior discussion of interacting QFTs is totally safe, and actually gives us a natural place for a width to appear.

Recall that we defined by $-iM^2(p^2)$ the sum of all 1PI diagrams:

$-iM^2(p^2) =$ 

and argued that the series of diagrams contributing to the propagator could be re-summed, so that:

 $= \dfrac{i}{p^2 - M^2(p^2)}$

(here I've shifted M to include the "bare mass")

We assumed the asymptotic states were defined to absorb these corrections, so that our discussion of S-matrix elements needed only the "amputated" diagrams.

As we saw now Feynman rules, the scattering amplitude of a process w/ an intermediate particle contains a ~~complex pole~~ pole in the propagator:

$$S \sim \frac{1}{p^2 - M^2}$$

Moving to the rest frame of the intermediate particle,

$$S \sim \frac{1}{p_0^2 - M^2} = \frac{1}{(p_0 + M)(p_0 - M)}$$

We can move to t-space via Fourier transform:

$$S \sim \int_{-\infty}^{\infty} dp^0 \frac{e^{-ip_0 t}}{(p_0 + M)(p_0 - M)}$$

Following our usual prescription, we close the contour in the lower half plane and pick up the pole at $p_0 = +M$:

Then,

$$S \sim e^{-iMt} = e^{-i\mathrm{Re}(Mt)} \; e^{\mathrm{Im}(Mt)}$$

$$\Rightarrow |S|^2 \sim e^{2\mathrm{Im}(Mt)}$$

$\Rightarrow$ the amplitude decays exponentially w/ a decay rate of $\Gamma = -2\mathrm{Im}(M)$

$\rightarrow$ Instabilities indicated by Imaginary part of the pole for a particle propagator (this is true in non-relativistic QM too).

This actually provides a formal, exact (non-perturbative) definition of the decay rate (inverse lifetime)

We define the 1PI corrections as $+i\Pi(p^2)$, then the "pole" of the full propagator is

$$M^2(p^2) = m_0^2 - \Pi(p^2)$$

evaluated at $p^2 = M^2$ (this is a recursive formula, which can be solved to arbitrary precision)

Then, the physical mass and decay rate are

$$m = \mathrm{Re}[M^2]$$

and $$\Gamma = \frac{1}{m}\mathrm{Im}\,\Pi(M^2)$$

Again, these are unambiguous, physical definitions of the mass and decay rate that manifest in experiments!

Now, near the resonance of an unstable particle, amplitudes have the form:

"Breit-Wigner form"

$$\frac{1}{p^2 - m^2 + im\Gamma} \overset{p^0 \simeq E_{\vec{p}}}{\simeq} \frac{1}{2E_{\vec{p}}\left(p^0 - E_{\vec{p}} + i\frac{m}{E_{\vec{p}}}\frac{\Gamma}{2}\right)}$$

This leads to physical processes such as cross sections,

$$\sigma \sim |M|^2 \sim \frac{1}{(E-E_0)^2 + \Gamma^2/4}$$

This leads to a peak in Energy near the particle's mass, with Γ governing the "width" of the peak.

For this reason, Γ is often called the "decay width" (I'll mostly use this terminology).

A few more comments on decay rates generally:

Usually, a particle will have many possible decay channels. We'll learn shortly how to calculate the decay rate Γ_i (proportional to the transition probability to the state i) individually.

The individual rates Γ_i are called "partial widths".

Starting from all of these being exponential decay, we see that, starting w/ N particles, the number that decay in the Δt is:

$$\Delta N = -N\Gamma_1 \Delta t - N\Gamma_2 \Delta t - \dots$$

$$= -N\left(\sum_i \Gamma_i\right)\Delta t \equiv -N\Gamma\Delta t$$

So the total decay rate (total width) is simply the sum of all the partial widths.

The particle's lifetime is $\tau = 1/\Gamma$

(note: $\Gamma \sim$ units of mass, $\tau \to$ units of sec)

An important quantity is the branching ratio of a particle into a chosen final state. This is just the relative frequency of a particular decay mode i:

$$Br_i \equiv \Gamma_i / \Gamma_{tot}$$

As an example, the SM Higgs boson decays into many final states: $b\bar{b}$, W^+W^-, ZZ, gg, $\gamma\gamma$, $Z\gamma$, Zl^+l^-, ...

Some of the branching ratios are:

$Br(h \to b\bar{b}) \simeq 58\%$

$Br(h \to \tau\tau) \simeq 0.063\%$

$Br(h \to \gamma\gamma) \simeq 0.0023\%$

(we'll discuss where these numbers come from next semester!)

Its total width is ~ 4 MeV, which is very small for its mass (in contrast, the Z boson, which is ~30% lighter has a width of ~2.5 GeV)

The Higgs lifetime is $\sim 1.6 \times 10^{-22}$ sec — even at the speed of light, $c\tau \sim 5 \cdot 10^{-14}$ m — you'll never actually "see" a Higgs boson, it decays too quickly!

Now lets turn to how we actually compute the decay width.

Recall a general transition amplitude was given by

$$\langle \vec{p}_1 \dots \vec{p}_n | iT | \vec{p}_A \vec{p}_B \dots \rangle = iM (2\pi)^4 \delta^4 \left(p_A + p_B + \dots - \sum_{f=1}^{n} \vec{p}_f \right)$$

where iM is given by the sum of fully connected, amputated Feynman diagrams.

From this, we can get the "transition probability":

$$P = \frac{|\langle \vec{p}_1 \dots \vec{p}_n | iT | \vec{p}_A \vec{p}_B \dots \rangle|^2}{\langle \vec{p}_1 \dots \vec{p}_n | \vec{p}_1 \dots \vec{p}_n \rangle \langle \vec{p}_A \vec{p}_B \dots | \vec{p}_A \vec{p}_B \dots \rangle}$$

where we've divided out the normalization of the in/out states.

The numerator takes the form:

$$|\langle \vec{p}_1 \dots \vec{p}_n | \overset{iT}{V} | \vec{p}_A \vec{p}_B \rangle|^2 = |M|^2 \left[(2\pi)^4 \delta^4 (p_A + p_B - \textstyle\sum_f p_f) \right]^2$$

$$= |M|^2 (2\pi)^4 \delta^4 (p_A + p_B - \textstyle\sum_f p_f) \cdot \underbrace{(2\pi)^4 \delta(0)}$$

$$= (2\pi)^4 \cdot \left(\int \frac{d^4x}{(2\pi)^4} e^{ip\cdot x} \Big|_{p=0} \right)$$

$$= \int d^4x = V \cdot T$$

The factors in the denominator follow from our normalization for one-particle states:

$$|\vec{p}\rangle = \sqrt{2E_{\vec{p}}}\, a^\dagger_{\vec{p}}\,|0\rangle$$

$$\Rightarrow \langle\vec{p}|\vec{p}\rangle = 2E_{\vec{p}}\langle 0|a_{\vec{p}}\,a^\dagger_{\vec{p}}|0\rangle = 2E_{\vec{p}}\,(2\pi)^3\,\delta^3(\vec{p}-\vec{p})$$

$$= 2E_{\vec{p}}\cdot V$$

So, for the $1\to n$ process of interest to us here,

$$\langle\vec{p}_A|\vec{p}_A\rangle = 2E_{\vec{p}_A}\cdot V$$

$$\langle\vec{p}_1\ldots\vec{p}_n|\vec{p}_1\ldots\vec{p}_n\rangle = 2E_1\ldots 2E_n\cdot V^n = \prod_{f=1}^{n}(2E_f\cdot V)$$

Then, dividing by T, we find the "transition probability per unit time":

$$\frac{P}{T} = \frac{1}{T}\cdot\frac{|\mathcal{M}|^2\,\delta^4(p_A-\sum_f p_f)\cdot V\cdot T}{(2E_A\cdot V)\left[\prod_f 2E_f\cdot V\right]}$$

$$= \frac{1}{2E_A}\cdot|\mathcal{M}|^2\,\delta^4(p_A-\sum_f p_f)\cdot\frac{1}{\left(\prod_f 2E_f\cdot V\right)}$$

This gives the transition probability into the particular momenta $\{p_f\}$.

To compute the total rate we want to sum over the possible momenta.

To do so, we convert P/T to a differential probability to have momenta in the intervals:

$$(\vec{p}_f, \vec{p}_f + d\vec{p}_f) \quad , \quad f = 1, \ldots n$$

This is done by multiplying P/T w/ a phase space factor, or density of states.

This is easily done in our box of spatial volume. The momenta are quantized:

$$\vec{p} = \frac{2\pi}{L}(n_x, n_y, n_z)$$

(box of side L w/ periodic b.c.'s)

$$\rightarrow d^3n = dn_x \cdot dn_y \cdot dn_z = \frac{L\,dp_x}{2\pi} \cdot \frac{L\,dp_y}{2\pi} \cdot \frac{L\,dp_z}{2\pi}$$

$$= V \cdot \frac{d^3p}{(2\pi)^3}$$

So we should multiply P/T by this factor for every final state particle:

$$\frac{dP}{T} = \frac{|M|^2}{2E_A}\,\delta^4(p_A - \Sigma p_f) \cdot \prod_f \frac{d^3p_f}{(2\pi)^3 \cdot 2E_f}$$

The decay rate for $\tilde{p}_A$ ~~the~~ is given by this partial probability in the A rest frame:

$$d\Gamma = \frac{dP}{T}\Big|_{E_A = m_A} = \frac{1}{2m_A}|\mathcal{M}|^2 \delta^4(p_A - \Sigma p_f) \times \prod_f \frac{d^3 p_f}{(2\pi)^3 2E_f}$$

and the partial width Γ_i is obtained by integrating over all final state momenta.

~~[illegible]~~

As an explicit example, consider a theory with two real scalar fields: Φ and ϕ, with masses M and m.

The Lagrangian is: $\mathcal{L} = \mathcal{L}_{free} + \mathcal{L}_{int}$ with

$$\mathcal{L}_{free} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2\phi^2 + \frac{1}{2}(\partial_\mu \Phi)^2 - \frac{1}{2}M^2\Phi^2$$

$$\mathcal{L}_{int} = -\frac{1}{2}A\,\Phi\,\phi\,\phi$$

We'll assume $M > 2m$. A is a coupling constant.

We want to compute the kinematically allowed $\Phi \to \phi\phi$ decay.

Let's first derive the Feynman rule for this process.
The interaction Hamiltonian is:

$$H_I(t) = \int d^4x \, \frac{1}{2} A \Phi_I \phi_I^2$$

We distinguish the external 1-particle states:

$$|\phi(p)\rangle = \sqrt{2E_{\vec{p}}}\, a^\dagger_{\vec{p}} |0\rangle$$

$$|\Phi(k)\rangle = \sqrt{2E_k}\, b^\dagger_{\vec{k}} |0\rangle$$

The T-matrix element relevant for the decay is:

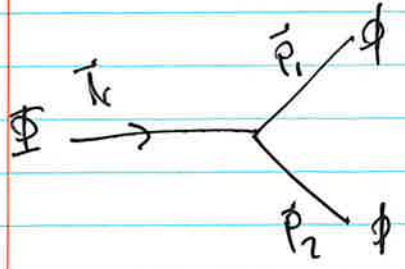
$$i \times 2 \times \langle \phi(p_1)\phi(p_2) | T\left\{ -i \int d^4x \, \frac{A}{2} \Phi \phi \phi \right\} | \Phi(k) \rangle$$

$$= -iA \frac{1}{2} \times 2 \times \int d^4x \, e^{-i(k - p_1 - p_2)\cdot x}$$

$$= -iA \cdot (2\pi)^4 \delta^4(k - p_1 - p_2)$$

The extra factor of 2 is because there are two equivalent ways of contracting the ϕ fields (hence our normalization of the coupling in $\mathcal{L}_{int}$)

This is the result from the Feynman diagram:



$= iA$ ($i \times \mathcal{L}_{int}$ divided by the fields, times the combinatorial factor)

This is an example of the general rule for scalars, that the Feynman rule for a vertex w/ $\phi_i, \phi_j, \ldots$ legs is given by

$$i \frac{\partial \mathcal{L}_{int}}{\partial \phi_i \ldots \partial \phi_j}$$

In this case, the $\Phi\phi\phi$ diagram gives us iA directly, so

$$iM = -iA \Rightarrow |M|^2 = A^2$$

Then,

$$d\Gamma = \frac{1}{2M} |M|^2 \delta^4(P - p_1 - p_2) \cdot \frac{d^3p_1}{(2\pi)^3} \cdot \frac{d^3p_2}{(2\pi)^3} \frac{1}{2E_1 2E_2}$$

For this special case of 2-body decays, we have:

$$E_1 = \frac{M^2 - m_2^2 + m_1^2}{2M}, \quad E_2 = \frac{M^2 - m_1^2 + m_2^2}{2M}$$

and $$|\vec{p}_1| = |\vec{p}_2| = \frac{1}{2M}\sqrt{\lambda(M^2, m_1^2, m_2^2)}$$

w/ $\lambda(a, b, c) = a^2 + b^2 + c^2 - 2ab - 2bc - 2ac$.

We can do the integral over d^3p_2 to absorb most of the δ-function,

then note that $d^3p_1 = |p_1|^2 d|p_1| d\Omega$,

w/ $d\Omega = d\phi_1 \cdot \sin\theta_1 d\cos\theta_1$, the angles of p_1 in the $\vec{K}$ rest frame.

These considerations give, for general $1 \to 2$ scattering w/ $m_1 \neq m_2$:

$$d\Gamma = \frac{1}{32\pi^2} |\mathcal{M}|^2 \frac{|\vec{p}_1|}{2M} d\Omega$$

Now, for our special case of $m_1 = m_2 = m$,

$$|\vec{p}_1| = \frac{1}{2M} \lambda^{1/2}(M^2, m^2, m^2) = \frac{1}{2M}\sqrt{M^4 - 4M^2m^2}$$

$$= \sqrt{\frac{M^2}{4} - m^2}$$

$\to$

~~$d\Gamma = \frac{1}{32\pi^2} A^2 \sqrt{\frac{M^2}{4} - m^2}\, d\Omega$~~

this includes a factor $1/2!$ for identical particles in the final state!

$$d\Gamma = \frac{1}{64\pi^2} A^2 \sqrt{\frac{1}{4} - \frac{m^2}{M^2}}\, d\Omega \qquad (4\pi)$$

$$= \frac{1}{32\pi} A^2 \cdot \frac{1}{M}\sqrt{1 - \frac{4m^2}{M^2}}$$

$$\simeq \frac{A^2}{32\pi M} \quad (m \ll M)$$

For e.g., $M \simeq 100\,\text{GeV}$, $A \approx 1\,\text{GeV}$, we find
$\Gamma \simeq 10^{-4}\,\text{GeV}$.

Assuming this is the only decay mode for $\mathbb{I}$, we find
$\tau = 1/\Gamma = 7 \times 10^{-21}\,\text{s}$

$\rightarrow$ $\mathbb{I}$ decays promptly.

Note that the kinematic factor $(1 - 4m^2/M^2)^{1/2}$ becomes real precisely above the threshold $M > 2m$.

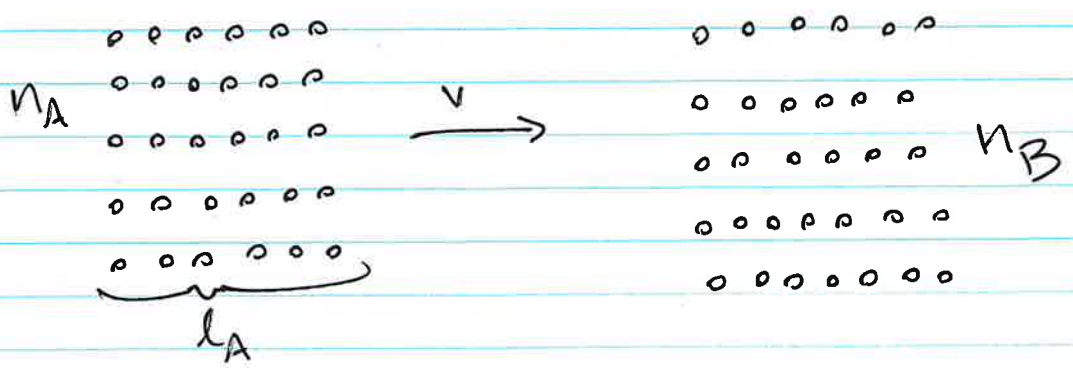
Scattering & the Cross Section

Now let's turn to 2-particle scattering. We want to consider a general experiment w/ two beams of particles w/ well-defined momenta are made to collide. We're assuming some idealized detector observes the products of the collision, to make inferences about the underlying process.

The probability of observing any particular final state is related to the so-called cross section.

The cross section is defined so that it depends only on the types of particles entering the collision, and their momenta. It does not depend, for instance, on the beam sizes or intensities.

To define the cross section, imagine a beam of particles A, w/ n_A particles per unit volume, and velocity v incident on a target of particles B at rest w/ number density n_B.



Suppose the beam / target have lengths ℓ_A, ℓ_B, resp.

The total number of scattering events (of any type) should be proportional to n_A, n_B, ℓ_A, ℓ_B and the cross-sectional area common to the two bunches'.

$$\#\text{ of scattering events} \equiv N = \sigma \times \left(n_A \ell_A \cdot n_B \ell_B \cdot A \right)$$

The cross-section σ is just the constant of proportionality here':

$$\sigma \equiv \frac{\#\text{ of scattering events}}{n_A \ell_A \cdot n_B \ell_B \cdot A}$$

Note that this definition is symmetric, so we could have taken A at rest, or any other frame related by a longitudinal boost.

It's clear from here that σ has units of area': it is essentially the effective area of the target particles that is "seen" by the beam.

If the experiment runs for some time T, such that N^A_{bunch} bunches of beam particles hit the target, then the total # of events is the product of the event rate and time.

Equivalently,

$$\text{rate of scattering events} \equiv R = \frac{N}{T} = \sigma \cdot \left[\left(\frac{N^A_{bunch} \cdot n_A \ell_A}{T} \right) \left(n_B \ell_B \cdot A \right) \right]$$

$$= \sigma \cdot \phi_A \cdot N_B$$

Where we've defined the <u>flux</u> of A particles,

$$\phi_A = N^A_{bunch} \cdot n_A \cdot \ell_A / T$$

and the total number of targets, $N_B = n_B \cdot \ell_B \cdot A$.

The flux is the # of particles crossing ~~the area~~ A unit area, in unit time.

The quantity $\phi_A * N_B$ is known as the <u>instantaneous luminosity</u>

$$\mathcal{L} = \phi_A * N_B$$

→ the number of crossings b/w the particles per unit time per unit area.

The total interaction rate is then

$$R = \mathcal{L} \times \sigma$$

Note that $\mathcal{L}$ is independent of the fundamental physics: it reflects experimental conditions, such as the beam size and intensity. σ contains the particle physics.

As an aside, note that the instantaneous luminosity is often expressed in different ways, depending on the type of experiment.

Physically, they're all related to just the number of crossings per unit time per unit area.

As an example, in a fixed target experiment: an intense beam impacts a target. If the # of incident particles A on target per unit time is Φ_A (do not confuse this w/ Φ_A above!) and the target is "thin" compared to the interaction length λ, then

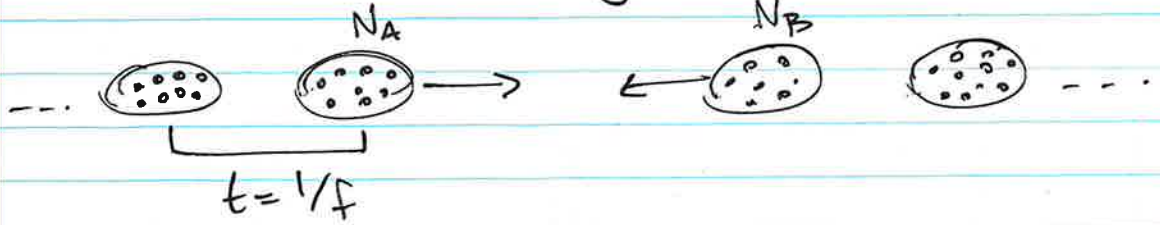
$$\mathcal{L} = \Phi_A \cdot n_B \cdot \ell_B \qquad (\ell_B \ll \lambda)$$

w/ ℓ_B the length of the target. If the target is "thick", w/ $\ell_B \gg \lambda$, the flux decreases w/ penetration distance z as an exponential. The instantaneous luminosity then becomes

$$\mathcal{L} = \Phi_A n_B \int_0^{\ell_B} dz \cdot e^{-z/\lambda} \simeq \Phi_A \cdot n_B \cdot \lambda$$

$$(\lambda \gg \ell_B)$$

Another example is a colliding beam experiment:



In this case, beams are made to intersect w/ each other. The beams consist of bunches w/ N_A ~~bunches~~ particles in each bunch for beam A, N_B for beam B. The bunches are separated by time t, corresponding to a collision frequency f. Then,

$$\mathcal{L} = f \frac{N_A \cdot N_B}{A}$$

w/ A the transverse cross-sectional area of the beams.

Given the instantaneous luminosity, we can integrate it over the run time of the experiment to find the integrated luminosity L:

$$L = \int dt \, \mathcal{L}$$

This gives the total number of crossings of incident particles & target particles per unit area.

Then, the total number of scattering events of a given type (computed w/ σ) is:

$$N = \sigma \times L$$

This is the key quantity of interest in a scattering experiment: we can predict the number of such events by calculating the cross section, and then compare it to the actual number of events observed.

Cross Section Units

The cross section has units of area (1/energy^2), so it is sometimes reported in units of cm^2.

Of course, the typical size of particle physics cross sections is much smaller, so its convenient to introduce the barn:

$$1 \text{ barn} \equiv 1\text{b} = 10^{-24} \text{ cm}^2$$

Even w/in particle physics, cross sections will vary over orders of magnitude, so we have prefixes:

$1 \text{ mb} = 10^{-3}$ b (millibarn) $= 10^{-27}$ cm^2

$1 \mu\text{b} = 10^{-6}$ b (microbarn) $= 10^{-30}$ cm^2

$1 \text{ nb} = 10^{-9}$ b (nanobarn) $= 10^{-33}$ cm^2

$1 \text{ pb} = 10^{-12}$ b (picobarn) $= 10^{-36}$ cm^2

$1 \text{ fb} = 10^{-15}$ b (femtobarn) $= 10^{-39}$ cm^2

The units of instantaneous luminosity are "per unit area per unit time", or

$$cm^{-2}\,s^{-1} \quad \text{or} \quad b^{-1}\,s^{-1}$$

"inverse barn per second".

The integrated luminosity, then is often expressed in units of b^{-1} (or pb^{-1}, fb^{-1}, ab^{-1}), which are particularly convenient in estimating statistics.

Let's consider some examples: the Large Hadron Collider (LHC) is designed to collide proton beams at a c.o.m energy of

$$E_{CM} = 2E_p = 14\ TeV \quad (1.4 \times 10^4\ GeV)$$

The design* instantaneous luminosity is

$$\mathcal{L} = 10^{34}\,cm^{-2}\,s^{-1} = 10\,nb^{-1}\,s^{-1}$$

In one year of running, this gives

$$L = \int dt \cdot \mathcal{L} = \frac{10\,nb^{-1}}{s} \cdot 1\,\cancel{yr}\left(\frac{3.14 \times 10^7\,s}{\cancel{yr}}\right)$$

$$\simeq 300\,fb^{-1}$$

(live time is usually closer to ~ 1/3 - 1/2 this)

TBC

Run 3 of the LHC (2022-2026) actually exceeded the design luminosity by a factor ~ 2

Since commissioning, it's accumulated ~ 550 fb^{-1}.

As of summer 2026, it's been shut down for upgrades to the "High-Luminosity"-LHC, which will achieve $\mathcal{L} \sim 5 \times 10^{34}\ cm^2/s$, and $L \simeq 4000\ fb^{-1}$ by ~2040.

The total proton-proton cross section measured during this run is

$$\sigma_{pp,tot} \simeq 111\ mb \quad \left(\begin{array}{c}\text{measured in Run 2} \\ @\ 13\ TeV\end{array}\right)$$

The total number of proton collision events is thus simply:

$$N_{pp} = \sigma_{pp,tot} \times L_{int}$$

$$= (111\ mb)(550\ fb^{-1})$$

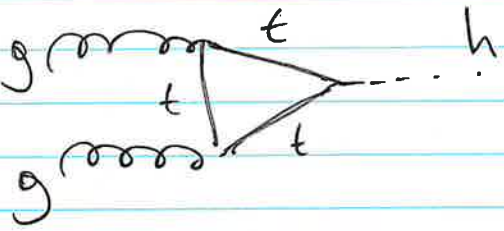
$$\simeq 5 \times 10^{16}\ \text{collisions}$$

The vast majority of these are totally uninteresting — a significant fraction are elastic collisions of the protons, and most are just rearrangements of the protons into other hadrons described by well-understood physics.

The more interesting processes involve e.g., W, Z and Higgs bosons or top quarks and these occur at a much lower rate.

As an example, the Higgs boson is produced predominantly via the "gluon fusion" process:

$g + g \to h$



The gluons are constituents of the proton — we'll eventually describe how this is calculated in the parton model.

The cross section for this process at 13 TeV is

$$\sigma(gg \to h) \simeq 50 \text{ pb}$$

So we can estimate the number of Higgs bosons produced so far:

$$N(gg \to h) = \sigma(gg \to h) \times L_{tot}$$

$$= (50 \text{ pb}) \times \left(\frac{550}{\text{fb}}\right)\left(\frac{10^3 \text{ fb}}{1 \text{ pb}}\right)$$

$$\simeq 27 \times 10^7$$

This is a large number — even for the "rare" decays such as $BR(h \to \gamma\gamma) \simeq 2 \times 10^{-3}$, we get

$$N(gg \to h \to \gamma\gamma) = \sigma(gg \to h) \cdot BR(h \to \gamma\gamma) \times L_{tot}$$

$$\simeq 2.7 \times 10^{7} \times 2 \times 10^{-3}$$

$$\simeq 60{,}000 \text{ events.}$$

This is a much more manageable number to analyze, but plenty to allow for robust statistical measurements of the Higgs properties.

In practice, these events must be "dug out" from backgrounds which have the same final states. This can be done by imposing cuts on the kinematic variables of the final state particles.

To actually compute a cross section, we follow a procedure similar to the decay width.

We first compute the transition amplitude (now for $2 \to n$ scattering):

$$\langle \vec{p}_1 \cdots \vec{p}_n | iT | \vec{p}_A \vec{p}_B \rangle = i\mathcal{M} (2\pi)^4 \delta^4 \left(p_A + p_B - \sum_f p_f \right)$$

and the transition probability per unit time:

$$\frac{P}{T} = \frac{1}{T} \frac{|\langle \vec{p}_1 \cdots \vec{p}_n | iT | \vec{p}_A \vec{p}_B \rangle|^2}{\langle \vec{p}_A \vec{p}_B | \vec{p}_A \vec{p}_B \rangle \langle \vec{p}_1 \cdots \vec{p}_n | \vec{p}_1 \cdots \vec{p}_n \rangle}$$

As before,

$$|\langle \vec{p}_1 \cdots \vec{p}_n | iT | \vec{p}_A \vec{p}_B \rangle|^2 = |\mathcal{M}|^2 (2\pi)^4 \delta^4 \left(p_A + p_B - \sum_f p_f \right) \times V \cdot T$$

while

$$\langle \vec{p}_A \vec{p}_B | \vec{p}_A \vec{p}_B \rangle = 2E_A \cdot 2E_B \cdot V^2$$

$$\langle \vec{p}_1 \cdots \vec{p}_n | \vec{p}_1 \cdots \vec{p}_n \rangle = \prod_f (2E_f \cdot V)$$

So,

$$\frac{P}{T} = \frac{1}{2E_A 2E_B \cdot V} \cdot \frac{1}{\prod_f 2E_f \cdot V} \cdot |\mathcal{M}|^2 (2\pi)^4 \delta^4(p_A + p_B - \textstyle\sum_f p_f)$$

Now we integrate over the final state phase space factors:

$$\frac{dP}{T} = \frac{|\mathcal{M}|^2 (2\pi)^4 \delta^4(p_A + p_B - \sum_f p_f)}{2E_A \cdot 2E_B \cdot V} \prod_f \frac{d^3 p_f}{(2\pi)^3 2E_f}$$

This is the rate for a single particle to scatter.
Recalling our earlier discussion:

$$R = \sigma \cdot \phi_A \cdot N_B = \sigma \cdot \phi_A \cdot 1 \quad \begin{pmatrix} \text{single} \\ \text{target particle} \end{pmatrix}$$

The flux of one particle in volume V, in a frame w/ A moving towards B w/ relative velocity $v_A - v_B$ is:

$$\phi_A = \frac{|v_A - v_B|}{V}$$



Then, equating $R = dP/T$, we simply divide by the flux to obtain the cross section:

$$d\sigma = \frac{1}{2E_A \cdot 2E_B |v_A - v_B|} |\mathcal{M}|^2 (2\pi)^4 \delta^4(p_A + p_B - \textstyle\sum_f p_f)$$

$$\times \prod_f \frac{d^3 p_f}{(2\pi)^3 2E_f}$$

This is the general formula for the differential cross section into momenta $(p_f, p_f + dp_f)$.

The factor

$$d\Pi_n \equiv \prod_{f=1}^{n} \frac{d^3 p_f}{(2\pi)^3 \cdot 2E_f} \times (2\pi)^4 \delta^4\left(p_A + p_B - \sum_f p_f\right)$$

is the "Lorentz-invariant phase space" for n final state particles. The 4-d δ-function and phase space factors are separately Lorentz invariant.

Note that if there are k identical particles in the final state, we must restrict the integration of $d\Pi_n$ to inequivalent configurations, or integrate over all final state momenta and divide by a factor $k!$.

The squared amplitude $|M|^2$ is also Lorentz-invariant. The prefactor remaining is known as the "flux factor"

$$F = 4 E_A \cdot E_B |v_A - v_B|$$

We can rewrite this in terms of Lorentz-invariant quantities. using $|v| = |\vec{p}|/E$,

$$F = 4 E_A E_B \cdot \left| \frac{|\vec{p}_A|}{E_A} + \frac{|\vec{p}_B|}{E_B} \right| = 4\left(E_A |\vec{p}_B| + E_B |\vec{p}_A|\right)$$

$$\Rightarrow F^2 = 16\left(E_A^2 |\vec{p}_B|^2 + E_B^2 |\vec{p}_A|^2 + 2 E_A \cdot E_B |\vec{p}_A| |\vec{p}_B|\right)$$

Now, note that

$$(p_A \cdot p_B)^2 = E_A^2 E_B^2 + |\vec{p}_A|^2 |\vec{p}_B|^2 + 2E_A \cdot E_B |\vec{p}_A||\vec{p}_B|$$

$$\Rightarrow F^2 = 16\left\{ E_A^2 |\vec{p}_B|^2 + E_B^2 |\vec{p}_A|^2 + \left[(p_A \cdot p_B)^2 - E_A^2 E_B^2 - |\vec{p}_A|^2 |\vec{p}_B|^2 \right] \right\}$$

$$= 16\left\{ (p_A \cdot p_B)^2 - (E_A^2 - |\vec{p}_A|^2)(E_B^2 - |\vec{p}_B|^2) \right\}$$

$$= 16\left[(p_A \cdot p_B)^2 - m_A^2 m_B^2 \right]$$

or,

$$F = 4\left[(p_A \cdot p_B)^2 - m_A^2 m_B^2 \right]^{1/2}$$

This is manifestly Lorentz invariant. Our final expression is compactly written as

$$d\sigma = \frac{1}{F} |\mathcal{M}|^2 d\Pi_n$$

w/

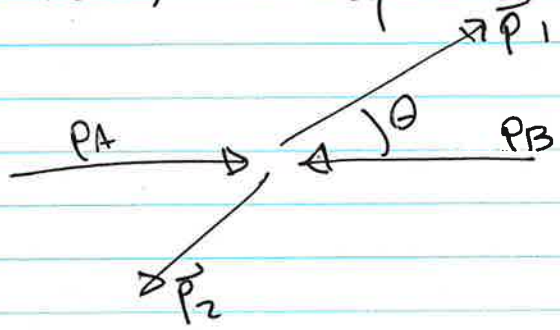
$$F = 4\left[(p_A \cdot p_B)^2 - m_A^2 m_B^2 \right]^{1/2}$$

$$d\Pi_n = (2\pi)^4 \delta^4\left(p_A + p_B - \textstyle\sum_f p_f \right) \prod_{f=1}^{n} \frac{d^3 p_f}{(2\pi)^3 2E_f}$$

2-to-2 Scattering

An important special case is 2-to-2 scattering. We will work in the CM frame, defined by

$$\vec{p}_A + \vec{p}_B = 0$$



The phase-space integral for 2-particle states simplifies enormously:

$$d\Pi_2 = (2\pi)^4 \delta^4(p_A + p_B - p_1 - p_2) \frac{d^3p_1}{(2\pi)^3 2E_1} \frac{d^3p_2}{(2\pi)^3 2E_2}$$

$$= \frac{1}{2^2 \cdot (2\pi)^2} \delta(E_A + E_B - E_1 - E_2) \cdot \frac{|\vec{p}_1|^2 \cdot d|\vec{p}_1| \, d\Omega}{E_1 \cdot E_2} \Bigg|_{\substack{\vec{p}_A = -\vec{p}_B \\ \vec{p}_1 = -\vec{p}_2}}$$

where $d\Omega = d\phi \, d\cos\theta$, ϕ, θ the azimuthal and polar angles of $\vec{p}_1$.

For the remaining δ-function, defining $E_{cm} = E_A + E_B$, and using

$$\delta(f(x)) = \sum_i \frac{1}{|f'(x_i)|} \delta(x - x_i)$$

$$\Rightarrow \delta\left(E_{cm} - \sqrt{|p_1|^2 + m_1^2} - \sqrt{|p_2|^2 + m_2^2}\right) = \frac{\delta\left(|\vec{p}_1| - |\vec{p}_1^{\,cm}|\right)}{\left(\frac{|p_1|}{E_1} + \frac{|\vec{p}_1|}{E_2}\right)\Big|_{\vec{p}_1 = \vec{p}_1^{\,cm}}}$$

w/

$$|\vec{p}_1|^{cm} = \frac{E_{cm}}{2} \lambda^{1/2}\left(1, \frac{m_1^2}{E_{cm}^2}, \frac{m_2^2}{E_{cm}^2}\right)$$

Then,

$$d\Pi_2 = \frac{1}{16\pi^2}\int \delta(|\vec{p}_1| - |\vec{p}_1^{\,cm}|)\,\frac{E_1 E_2}{(E_1+E_2)|\vec{p}_1|}\cdot\frac{|\vec{p}_1|^2 d|\vec{p}_1| d\Omega}{E_1 E_2}$$

$$= \frac{1}{16\pi^2}\frac{|\vec{p}_1^{\,cm}|}{E_{cm}}d\Omega$$

Plugging this into our general cross-section formula,

$$\sigma = \frac{1}{4E_A E_B}\frac{1}{|v_A - v_B|}\frac{1}{16\pi^2}\frac{|\vec{p}_1^{\,cm}|}{E_{cm}}\int |\mathcal{M}|^2 d\Omega$$

$$= \frac{1}{4E_A E_B |v_A - v_B|}\frac{1}{8\pi^2}\lambda^{1/2}\left(1, \frac{m_1^2}{E_{cm}^2}, \frac{m_2^2}{E_{cm}^2}\right)\int |\mathcal{M}|^2 \frac{d\Omega}{4\pi}$$

This may also be written as the differential cross section:

$$\left(\frac{d\sigma}{d\Omega}\right)_{cm} = \frac{1}{4E_A E_B}\frac{1}{|v_A - v_B|}\frac{1}{16\pi^2}\frac{|\vec{p}_1^{\,cm}|}{E_{cm}}|\mathcal{M}|^2$$

In the special case that all four particles have the same masses (e.g., $m \to 0$), then:

$$E_A = E_B = E_1 = E_2 = E_{cm}/2 \quad , \quad |v_A - v_B| = 2v_A = 2v_1 = \frac{2|\vec{p}_1^{\,cm}|}{E_{cm}/2}$$

$$\Rightarrow \left(\frac{d\sigma}{d\Omega}\right)_{cm} = \frac{1}{4(E_{cm}^2/4)}\frac{E_{cm}}{4|\vec{p}_1^{\,cm}|}\frac{1}{16\pi^2}\frac{|\vec{p}_1^{\,cm}|}{E_{cm}}|\mathcal{M}|^2$$

$$= \frac{1}{64\pi^2 E_{cm}^2}|\mathcal{M}|^2$$

The units of cross section are area, or in natural units, GeV^{-2}. The conversion to more common units of measurement is

$$\frac{1}{GeV^2} = \frac{1}{GeV^2}\left(\frac{1\,GeV}{1.602\times10^{-10}\,J}\right)^2\left(\frac{1.055\times10^{-34}\,J\cdot s}{\hbar}\right)^2$$

$$\times\left(\frac{3\times10^{10}\,cm/s}{c}\right)^2$$

$$= 3.9\times10^{-28}\,cm^2$$

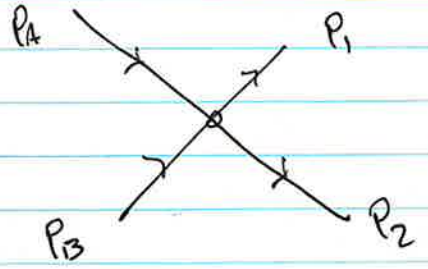
$$= 3.9\times10^{-4}\,b$$

Scott

Scattering Examples

ϕ^4 Theory:

Earlier we calculated the scattering amplitude for $\phi(p_A) + \phi(p_B) \to \phi(p_1) + \phi(p_2)$:



$$\to \quad i\mathcal{M} = -i\lambda$$

Since all ϕ particles have the same mass, the differential cross section is simply:

$$\left.\frac{d\sigma}{d\Omega}\right|_{cm} = \frac{1}{64\pi^2 E_{cm}^2}|\mathcal{M}|^2 = \frac{\lambda^2}{64\pi^2 E_{cm}^2}$$

The differential cross section has no non-trivial dependence on the angles, so we can simply integrate:

$$\int d\Omega = \int_0^{2\pi} d\phi \int_{-1}^{1} d\cos\theta = 4\pi$$

We must also divide by $2!$ to account for the identical particles in the final state. We find:

$$\sigma = \frac{1}{2!}\int d\Omega \frac{d\sigma}{d\Omega} = \frac{\lambda^2}{32\pi E_{cm}^2}$$

For e.g., $\lambda = 0.1$, $E_{cm} = 14$ TeV: $\sigma \simeq 5 \times 10^{-13}\,\text{GeV}^{-2} = 0.2$ fb

Scattering in $\Phi\phi\phi$ theory:

An example with much more interesting structure comes from the cubic interaction we discussed earlier:

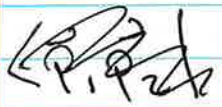
$$\mathcal{L} \supset -\frac{1}{2}M^2\Phi^2 - \frac{1}{2}m^2\phi^2 - \frac{1}{2}A\Phi\phi^2$$

Unlike our decay example, we won't make any assumptions about M vs. m for now.

One more time, let's look at this scattering process via the matrix element:

$$H_I = \int d^4x \, \frac{1}{2}A\Phi_I\phi_I^2$$

~~So we are interested in:~~ We're going to compute



$$\phi(p_A)+\phi(p_B) \to \phi(p_1)+\phi(p_2)$$

But now, there are only 2 ϕ_I fields in the vertex. To obtain a $\neq 0$ contribution to 2-to-2 scattering, we must go to 2nd order in the coupling A.

We're interested in

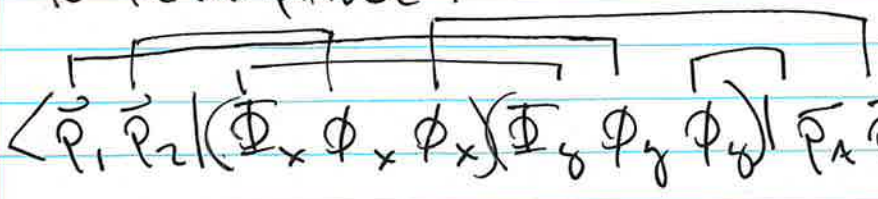
$$\langle \vec{p}_1 \vec{p}_2 | T\left\{ \frac{1}{2!}\left[-i\int d^4x \frac{A}{2}\Phi_x \phi_x \phi_x\right]\left[-i\int d^4y \frac{A}{2}\Phi_y \phi_y \phi_y\right]\right\} | \vec{p}_A \vec{p}_B \rangle$$

There are a number of ways these contractions can be performed.

In every case, we must have $\overbrace{\Phi_x \quad \Phi_y} = D_F^{(\Phi)}(x-y)$

But then we could have the ϕ_x contract w/ the initial states & ϕ_y w/ the final states, or equivalently, the opposite. Each of these has two equivalent choices for which ϕ contracts w/ which external state.

We could also have one ϕ_x and one ϕ_y contract w/ the initial states, and similarly for the final states. This leads to a totally different contribution to the amplitude:



$$\langle \vec{p}_1 \vec{p}_2 | (\Phi_x \phi_x \phi_x)(\Phi_y \phi_y \phi_y) | \vec{p}_A \vec{p}_B \rangle$$

$$\sim e^{-iq\cdot(x-y)} \cdot e^{-i(p_B+p_2)\cdot x} \, e^{+i(p_A-p_1)\cdot y}$$

vs.

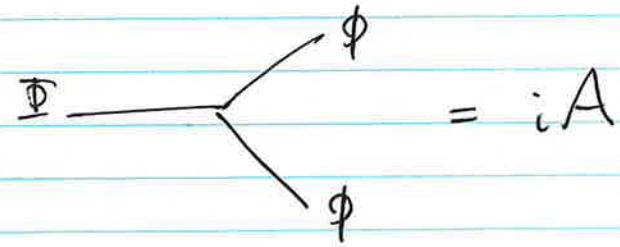
$$\langle \vec{p}_1 \vec{p}_2 | (\Phi_x \phi_x \phi_x)(\Phi_y \phi_y \phi_y) | \vec{p}_A \vec{p}_B \rangle$$

$$\sim e^{-iq\cdot(x-y)} \; e^{+i(p_1+p_2)\cdot x} \; e^{-i(p_A+p_B)\cdot y}$$

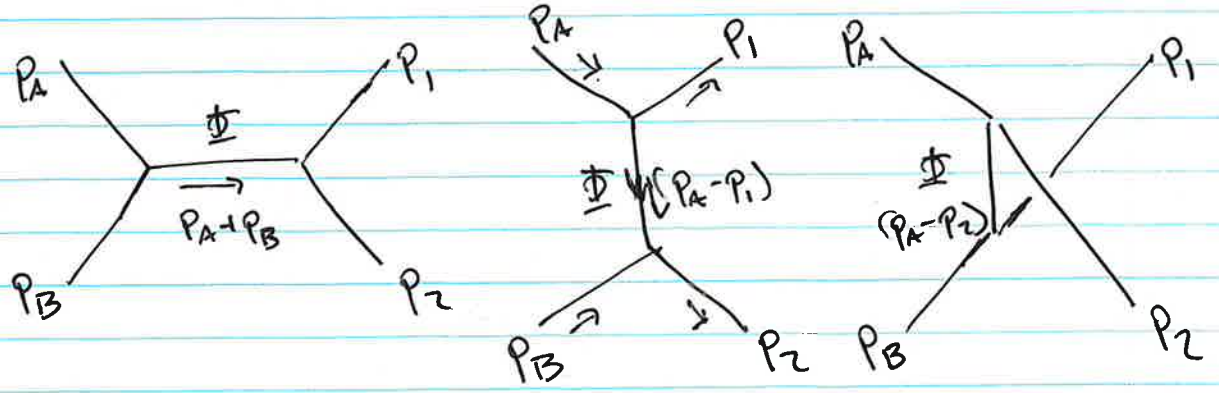
These will yield different structured S-functions conserving momentum.

Rather than work these out, it's easy to see how these possibilities manifest in Feynman diagrams:

We already found the vertex rule:



There are 3 distinct possibilities for 2-to-2 scattering at $O(A^2)$:



(1) (2) (3)

Working directly in momentum space:

$$(1): \quad i\mathcal{M}_s = (iA)\frac{i}{(p_A+p_B)^2 - M^2}(iA)$$

$$(2): \quad i\mathcal{M}_t = (iA)\frac{i}{(p_A-p_1)^2 - M^2}(iA)$$

$$(3): \quad i\mathcal{M}_u = (iA)\frac{i}{(p_A-p_2)^2 - M^2}(iA)$$

The only difference is the momentum appearing in the propagator. We've labelled these amplitudes suggestively w/ an s, t and u. These correspond to the so-called Mandelstam variables: three Lorentz-invariant, scalar products that appear in 2-to-2 scattering.

Starting from total 4-momentum conservation:

$$p_A^\mu + p_B^\mu = p_1^\mu + p_2^\mu$$

We can rearrange this in 3 distinct ways, and square the quantities. These are:

$$s = (p_A+p_B)^2 = (p_1+p_2)^2$$

$$t = (p_A-p_1)^2 = (p_2-p_B)^2$$

$$u = (p_A-p_2)^2 = (p_1-p_B)^2$$

Note that s is simply the CM energy squared in the CM frame:

$$p_A^\mu = (E_A, \vec{p}) \quad , \quad p_B^\mu = (E_B, -\vec{p})$$

$$\rightarrow (p_A + p_B)^\mu = (E_A + E_B, \vec{0}) = (E_{CM}, \vec{0})$$

$$\rightarrow s = (p_A + p_B)^2 = E_{CM}^2$$

The Mandelstam variables also satisfy

$$s + t + u = (p_A + p_B)^2 + (p_A - p_1)^2 + (p_A - p_2)^2$$

$$= (m_A^2 + m_B^2 + 2p_A \cdot p_B) + (m_A^2 + m_1^2 - 2p_A \cdot p_1)$$

$$+ (m_A^2 + m_2^2 - 2p_A \cdot p_2)$$

$$= 3m_A^2 + m_B^2 + m_1^2 + m_2^2 - 2p_A \cdot (p_B - p_1 - p_2)$$

$$= m_A^2 + m_B^2 + m_1^2 + m_2^2$$

where $p_B - p_1 - p_2 = p_A$, so $-2p_A \cdot (p_B - p_1 - p_2) = -2p_A^2 = -2m_A^2$.

The sum of $s+t+u$ is equal to the sum of all the squared masses. This can be useful in simplifying the final expressions.

In terms of s, t, u the amplitudes take the form:

$$\mathcal{M}_s = \frac{-A^2}{s-M^2}, \quad \mathcal{M}_t = \frac{-A^2}{t-M^2}, \quad \mathcal{M}_u = \frac{-A^2}{u-M^2}$$

So the full amplitude is

$$\mathcal{M} = \mathcal{M}_s + \mathcal{M}_t + \mathcal{M}_u$$
$$= -A^2 \left[\frac{1}{s-M^2} + \frac{1}{t-M^2} + \frac{1}{u-M^2} \right]$$

The individual amplitudes are often refered to as "s-channel", "t-channel" and "u-channel", respectively.

Lets press on and work out the kinematics in detail.

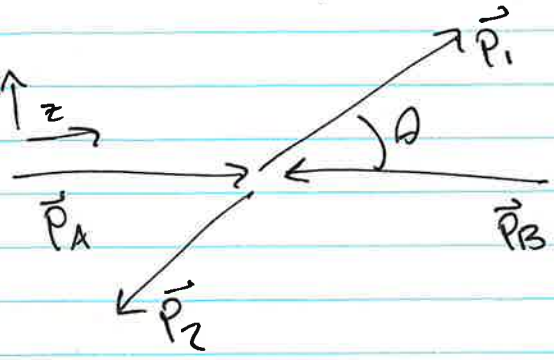
Sliding to the CM frame: $\vec{p}_A + \vec{p}_B = 0 \Rightarrow |\vec{p}_A| = |\vec{p}_B| = p$

and $E_A = E_B = E_1 = E_2 = E \equiv \sqrt{p^2 + m^2}$

(since all 4 ϕ's have the same mass)

All four particles lie in a single plane, which we take to be the x-z plane, w/ the initial momenta along the z-direction.

In this frame, the four-momenta can be parameterized explicitly as:



$$\vec{p}_A = (E, 0, 0, p)$$

$$\vec{p}_B = (E, 0, 0, -p)$$

$$\vec{p}_1 = (E, p\sin\theta, 0, p\cos\theta)$$

$$\vec{p}_2 = (E, -p\sin\theta, 0, -p\cos\theta)$$

The Mandelstam variables become:

$$s = (p_A + p_B)^2 = 4E^2 = E_{CM}^2$$

$$t = (p_A - p_1)^2 = p_A^2 + p_1^2 - 2p_A \cdot p_1$$

$$= 2m^2 - 2(E^2 - p^2\cos\theta)$$

Now we can write, E, p in terms of s, m:

$$E^2 = s/4$$

$$p^2 = E^2 - m^2 = \frac{s}{4} - m^2 = \frac{s}{4}\left(1 - \frac{4m^2}{s}\right) \equiv \frac{s}{4}\beta^2$$

Where we have defined $\beta = (1 - 4m^2/s)^{1/2}$. Note that $\beta = p/E$ is simply the velocity of the particles in the CM frame (Hence the choice of name!)

Then, we have:

$$t = 2m^2 - \frac{s}{2}(1-\beta^2\cos\theta)$$

$$u = (p_A - p_2)^2 = p_A^2 + p_2^2 - 2p_A\cdot p_2 = 2m^2 - 2(E^2 + p^2\cos\theta)$$

$$= 2m^2 - \frac{s}{2}(1+\beta\cos\theta)$$

The squared amplitude is:

$$|\mathcal{M}|^2 = A^4\left[\frac{1}{s-M^2} + \frac{1}{2m^2-M^2-\frac{s}{2}(1-\beta^2\cos\theta)} + \frac{1}{2m^2-M^2-\frac{s}{2}(1+\beta^2\cos\theta)}\right]^2$$

This has a non-trivial dependence on $\cos\theta$. Recalling our formula above for the differential cross section in the CM frame w/ all equal masses, we have

$$\frac{d\sigma}{d\Omega} = \frac{|\mathcal{M}|^2}{64\pi^2 E_{cm}^2}$$

We can integrate over the azimuthal angle & replace $E_{cm}^2 = s$, to find

$$\frac{d\sigma}{d\cos\theta} = \frac{A^4}{32\pi s}\left[\frac{1}{s-M^2} + \frac{1}{2m^2-M^2-\frac{s}{2}(1-\beta^2\cos\theta)} + \frac{1}{2m^2-M^2-\frac{s}{2}(1+\beta^2\cos\theta)}\right]^2$$

There are a number of limiting regimes. First, take $M \simeq m$, so that $2m^2 - M^2 \simeq m^2$. Then, we can look at the near threshold region: $s = 4E^2 = 4m^2\left(1+\frac{p^2}{m^2}\right)$

$\geq 4m^2$

i.e., $\beta^2 = p^2/m^2 \ll 1$ (the non-relativistic regime)

Then,

$$\frac{d\sigma}{d\cos\theta} = \frac{A^4}{32\pi(4m^2)}\left[\frac{1}{4m^2-m^2} + \frac{1}{m^2-4m^2/2} + \frac{1}{m^2-4m^2/2}\right]$$

$$\simeq \frac{A^4}{m^6} \longrightarrow \text{const., indep. of } \cos\theta$$

A similar thing happens in the $M \to 0$ limit, near threshold as well.

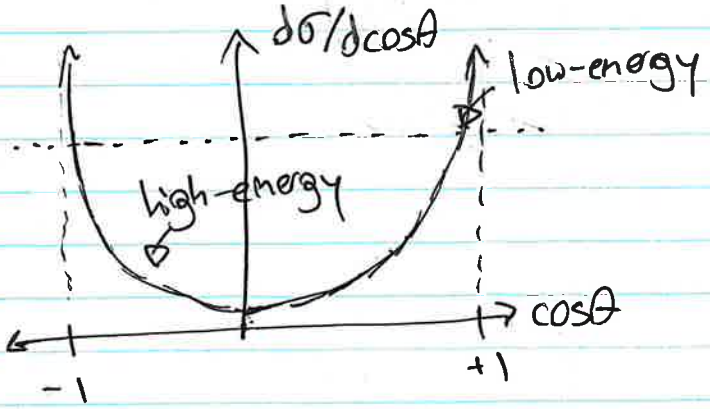
We can also take the high-energy limit: $s \gg m^2, M^2$
i.e., $\beta \simeq 1$:

$$\frac{d\sigma}{d\cos\theta} \to \frac{A^4}{32\pi s}\left[\frac{1}{s} - \frac{2}{s(1-\cos\theta)} - \frac{2}{s(1+\cos\theta)}\right]^2$$

This yields singularities in the forward ($\cos\theta \to 1$) and backward ($\cos\theta \to -1$) directions. These are cut off by m^2/s factors that we've dropped.

This is characteristic of t- and u-channel processes in the high-energy limit.

These two limits can be sketched as:



There is also the limit $M^2 \gg m^2$:

For $s \ll M^2$, we have

$$\frac{d\sigma}{d\cos\theta} \to \frac{A^4}{32\pi s}\left(\frac{-3}{M^2}\right)^2 = \frac{9A^4}{32\pi M^4}\cdot\frac{1}{s}$$

which is again a constant, independent of $\cos\theta$, but falls very rapidly with large M.

We can also have $s \simeq M^2 \gg m^2$.

This case actually requires treating the Feynman rules more carefully: since Φ is unstable, its propagator is

$$\xrightarrow[p]{\Phi} = \frac{i}{(p^2 - M^2 + iM\Gamma_\Phi)}$$

w/ Γ_Φ the total width we computed earlier:

$$\Gamma_\Phi \simeq \frac{A^2}{32\pi}\frac{1}{M} \qquad (M \gg m)$$

For small $\delta = s - M^2$, the s-channel amplitude ~~takes the form~~ leads to the Breit-Wigner form:

$$\frac{d\sigma}{d\cos\theta} \to \frac{A^4}{32\pi s}\left(\frac{1}{(s-M^2)+M^2\Gamma^2}\right)$$

In any of these limits, we can integrate over the polar angle to obtain the total cross section:

$$\sigma = \frac{1}{2!}\frac{A^4}{32\pi s}\int_{-1}^{1} d\cos\theta \left[\frac{1}{s-M^2} + \frac{1}{2m^2-M^2-\frac{s}{2}(1-\beta^2\cos\theta)} + \frac{1}{2m^2-M^2-\frac{s}{2}(1+\beta^2\cos\theta)}\right]^2$$

The factor $1/2!$ is for the two identical ϕ particles in the final state.

This is (at least numerically) straightforward, but not particularly illuminating.